



FATIMA JINNAH WOMEN UNIVERSITY

Lab 01

GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

Name: Saba Gull

Registration No: 2023-BSCS-074

Course Code: CS-649 (Cloud DevOps)

Section: A

GitHub Username: sabagu003-ui

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Task 1: Create and Configure Your GitHub Account

Step 1: Created my GitHub account with the username sabagu003-ui and opened my profile page.

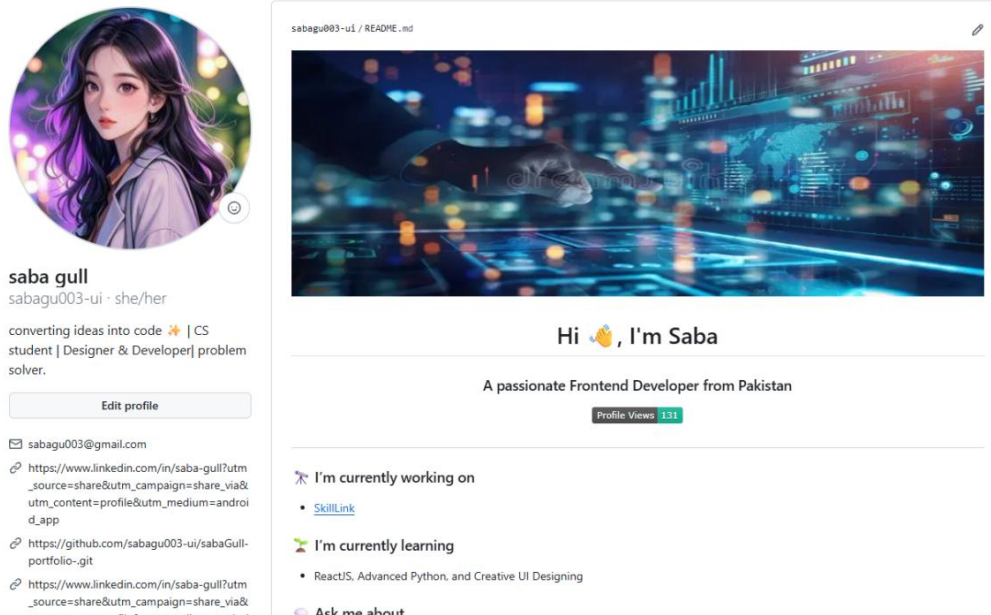


Figure 1.1: GitHub profile showing my profile name and GitHub username.

Step 2: Opened GitHub > Settings > Public profile and entered my name in the Name field.

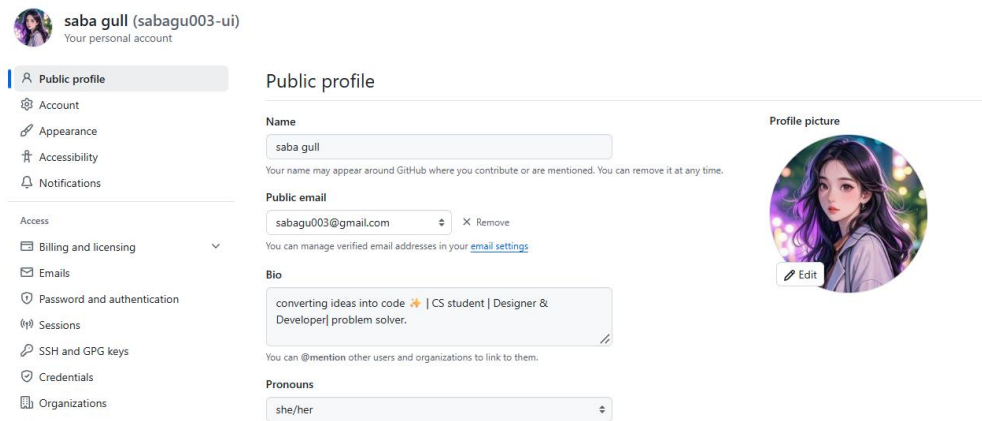


Figure 1.2: GitHub Public profile settings showing my name in the Name field.

Task 2: Register in the Student Marks Portal

Step 1: Opened the Student Marks Portal login page.

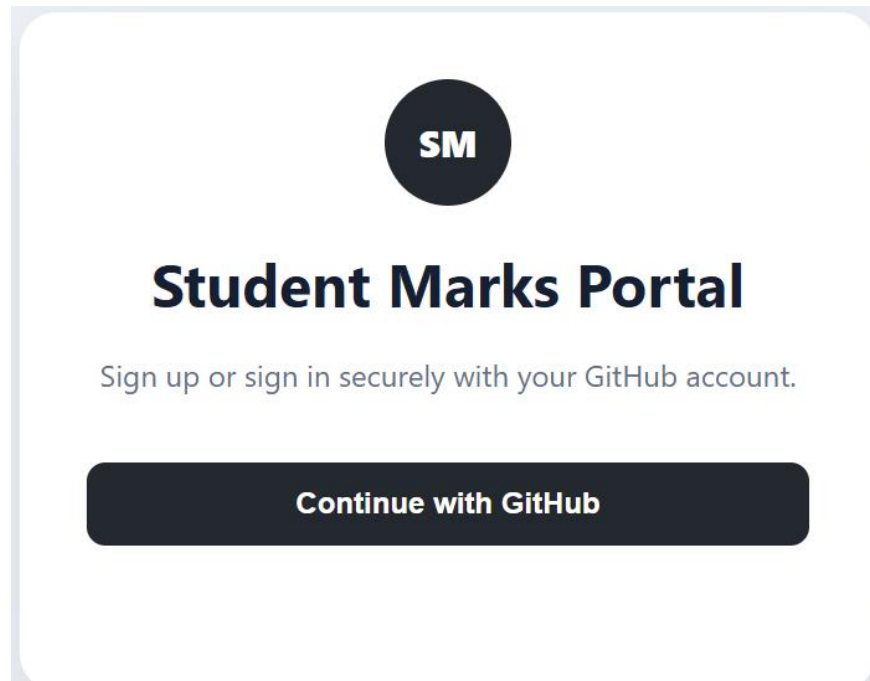


Figure 2.1: Student Marks Portal login page with the Continue with GitHub button.

Step 2: Selected Continue with GitHub and authorized the portal with my GitHub account.

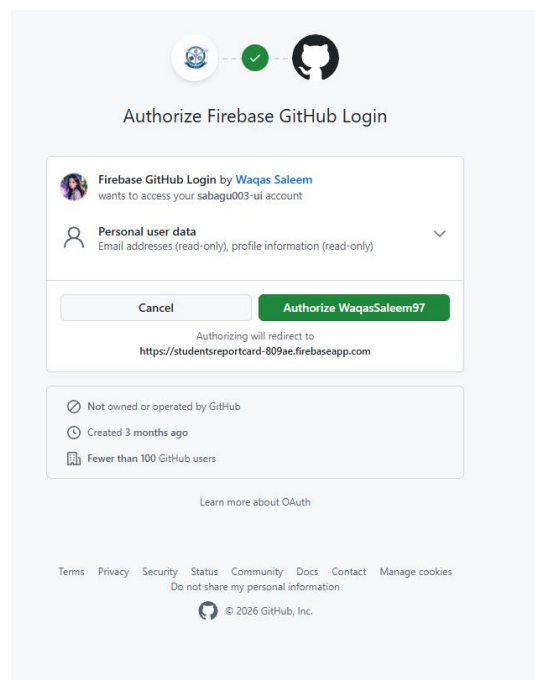
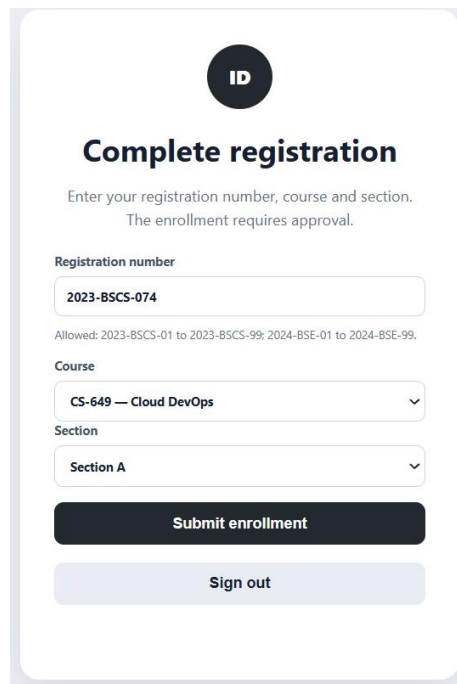


Figure 2.2: GitHub authorization screen for the Student Marks Portal.

Step 3: Entered my registration number, selected the course and section, and submitted the enrollment.



The form is titled "Complete registration" and includes a sub-header "Enter your registration number, course and section. The enrollment requires approval." It contains three input fields: "Registration number" with the value "2023-BSCS-074", "Course" with the value "CS-649 — Cloud DevOps", and "Section" with the value "Section A". Below these fields are two buttons: "Submit enrollment" and "Sign out".

ID

Complete registration

Enter your registration number, course and section.
The enrollment requires approval.

Registration number

2023-BSCS-074

Allowed: 2023-BSCS-01 to 2023-BSCS-99; 2024-BSE-01 to 2024-BSE-99.

Course

CS-649 — Cloud DevOps

Section

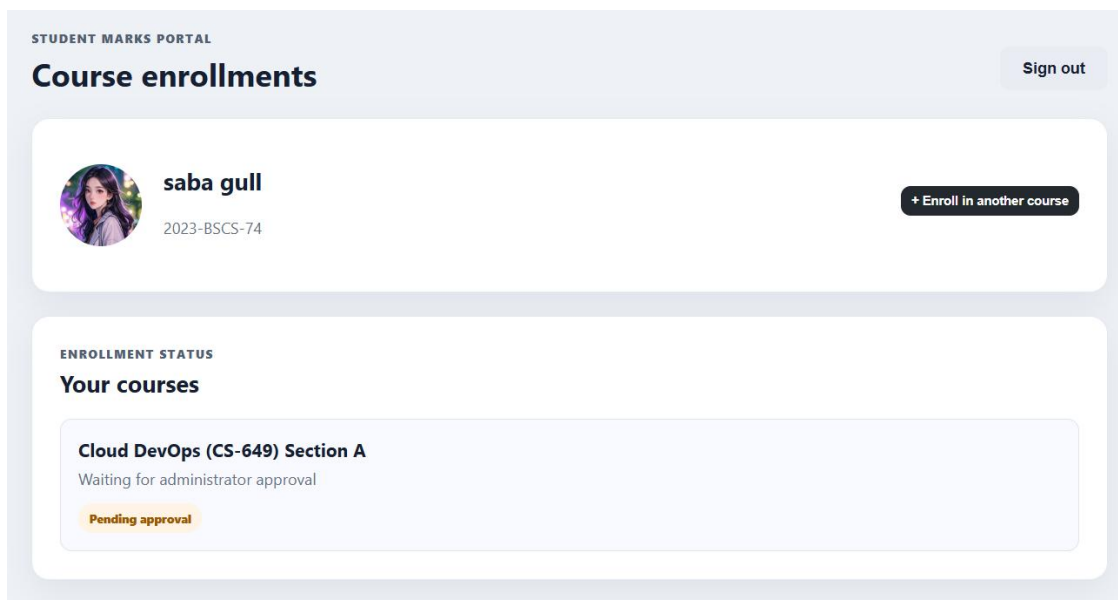
Section A

Submit enrollment

Sign out

Figure 2.3: Complete registration form with registration number, course CS-649 and Section A.

Step 4: Checked the portal after submission to confirm my profile and enrollment status.



The page is titled "STUDENT MARKS PORTAL" and "Course enrollments". It shows a user profile for "saba gull" with the registration number "2023-BSCS-74". Below the profile, there is a section titled "ENROLLMENT STATUS" and "Your courses". The course listed is "Cloud DevOps (CS-649) Section A" with the status "Waiting for administrator approval" and a "Pending approval" label. A button "+ Enroll in another course" is also visible.

STUDENT MARKS PORTAL

Course enrollments

Sign out

 **saba gull**
2023-BSCS-74

+ Enroll in another course

ENROLLMENT STATUS

Your courses

Cloud DevOps (CS-649) Section A
Waiting for administrator approval
Pending approval

Figure 2.4: My GitHub profile information and the pending enrollment in the portal.

Task 3: Download VMware Workstation Pro and Ubuntu Server

Step 1: Downloaded and installed VMware Workstation Pro on my Windows computer.

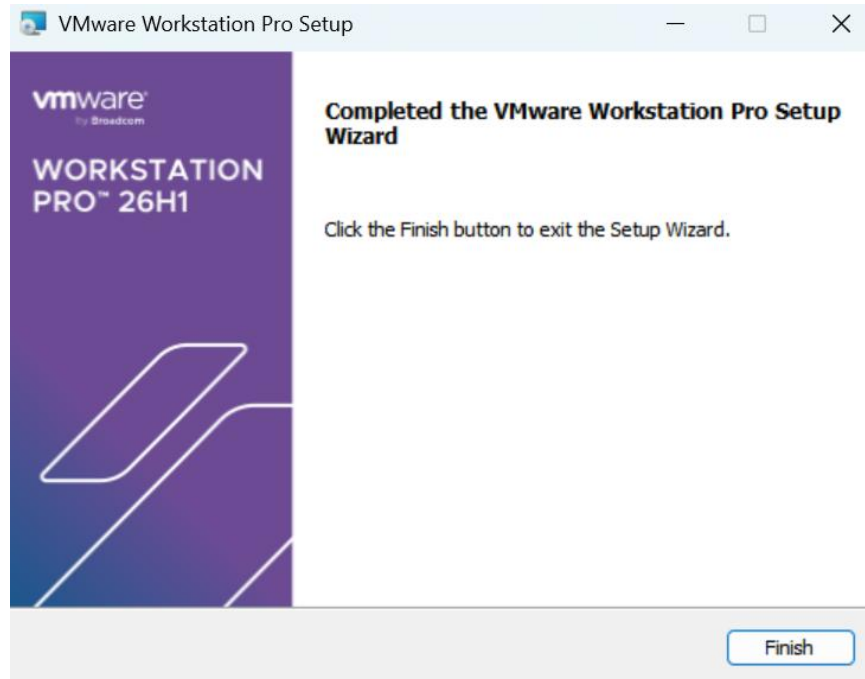


Figure 3.1: VMware Workstation Pro setup completed successfully.

Step 2: Downloaded the latest Ubuntu Server LTS ISO from the Ubuntu website.

Name	Date modified	Type	Size
▼ Today			
 VMware-Workstation-Full-26H1u1-25688693	04/10/2026 1:32 pm	Application	274,033 KB
 ubuntu-26.04.1-live-server-amd64	04/10/2026 2:15 pm	Disc Image File	2,859,240 KB

Figure 3.2: Downloads folder showing the Ubuntu Server ISO file and the VMware installer.

Step 3: Checked that Drive C: has more than 20 GB of free storage.

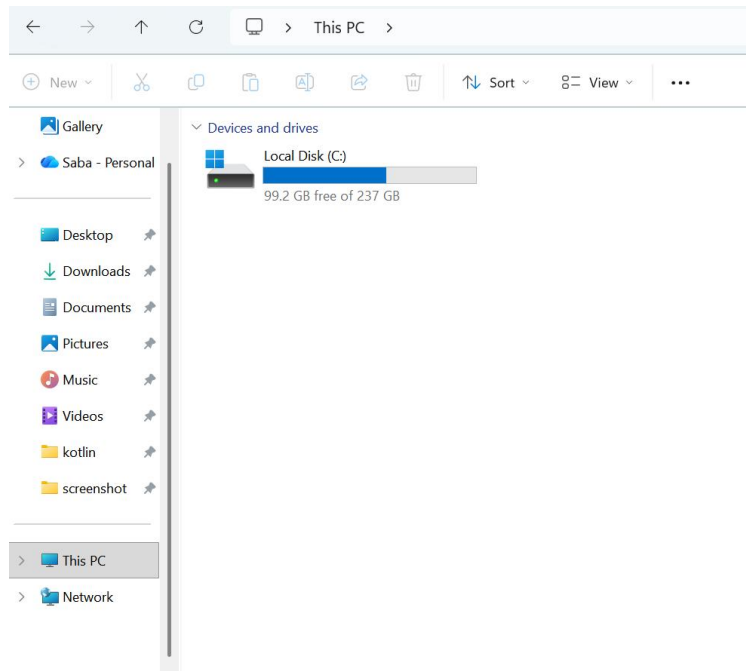


Figure 3.3: This PC showing 99.2 GB free on Local Disk (C:).

Task 4: Create the Ubuntu Server Virtual Machine

Step 1: Opened VMware Workstation Pro and selected Create a New Virtual Machine.

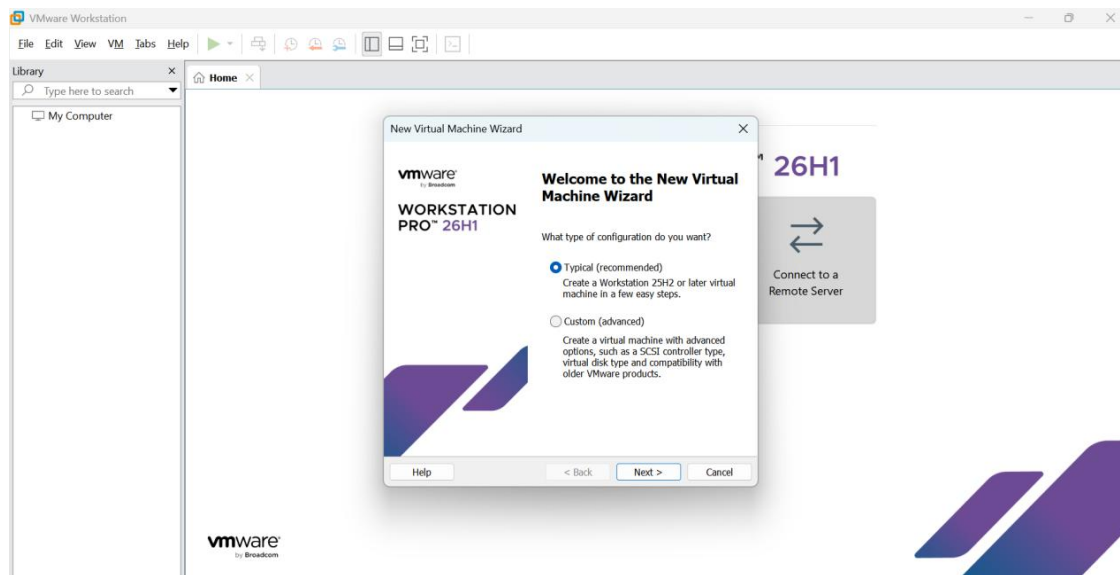


Figure 4.1: New Virtual Machine Wizard opened in VMware Workstation Pro.

Step 2: Selected the Typical (recommended) configuration.



Figure 4.2: Typical (recommended) configuration selected.

Step 3: Selected Installer disc image file (ISO) and browsed to the Ubuntu Server ISO.

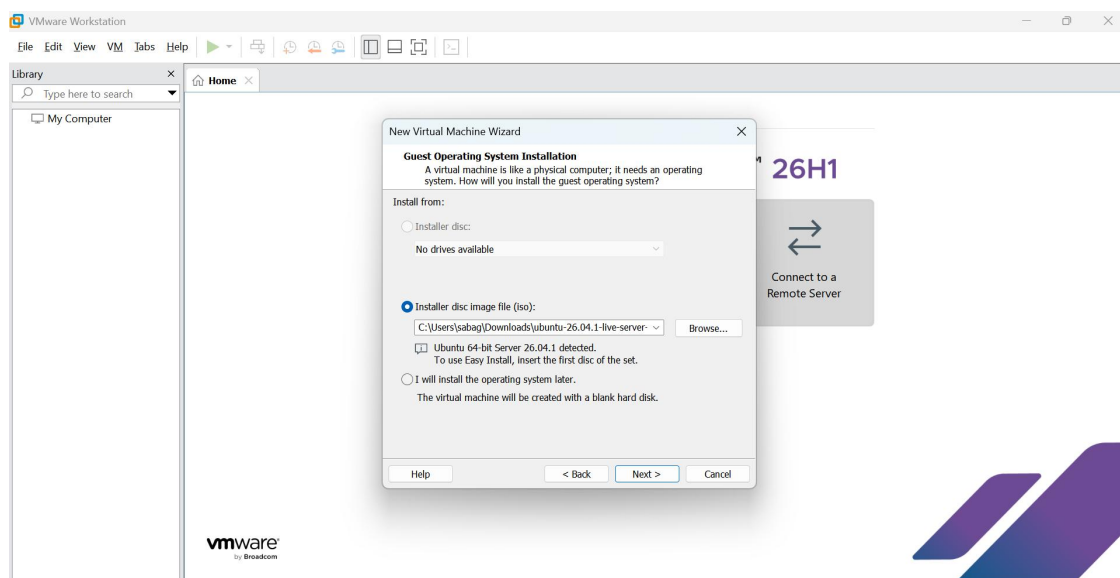


Figure 4.3: Ubuntu Server ISO selected and detected by the wizard.

Step 4: Allocated 25 GB of virtual disk space and stored it as a single file.

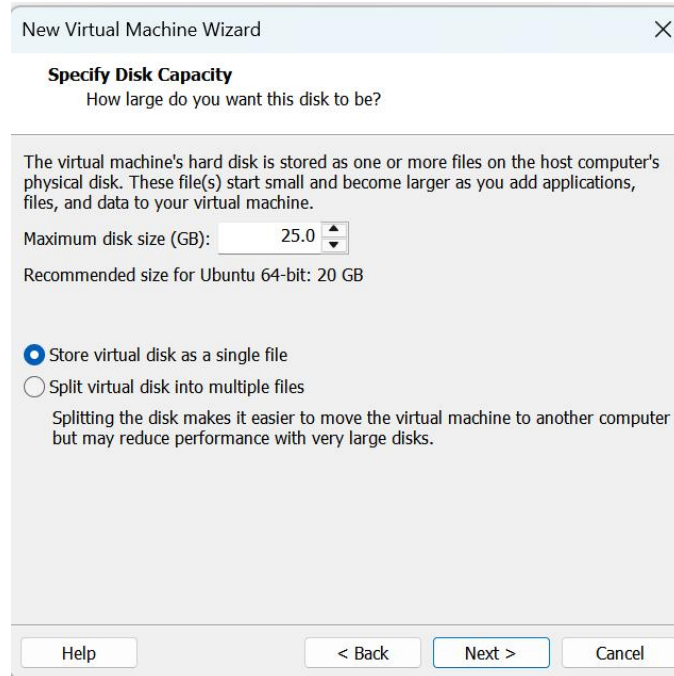


Figure 4.4: Maximum disk size set to 25 GB.

Task 5: Choose the Installation Language

Step 1: Used the arrow keys to select English as the installation language.

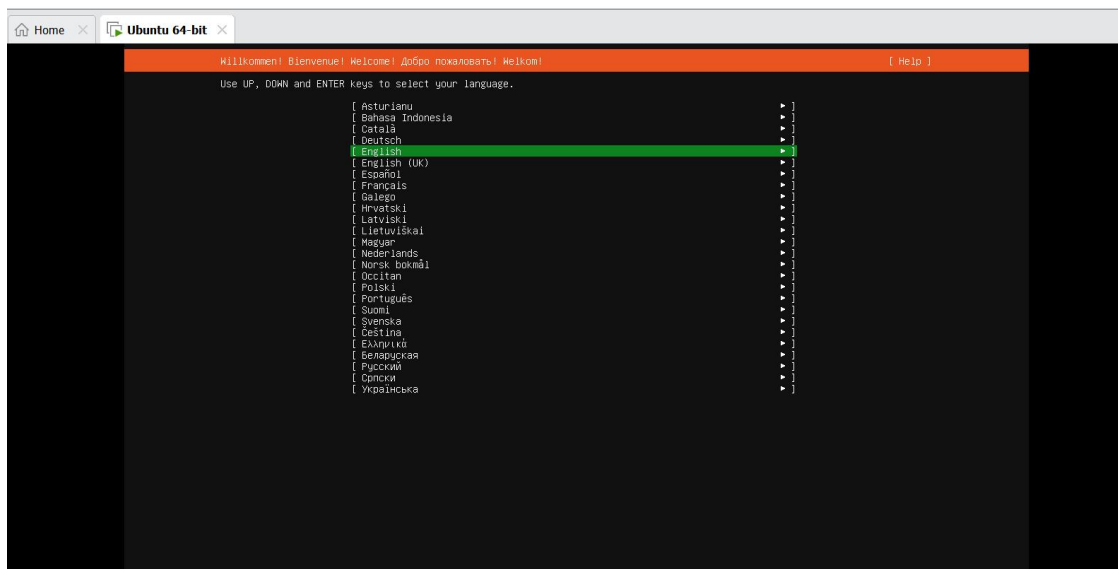


Figure 5.1: English selected on the language screen.

Task 6: Configure the Keyboard Layout

Step 1: Selected the English (US) keyboard layout and variant, then continued with Done.

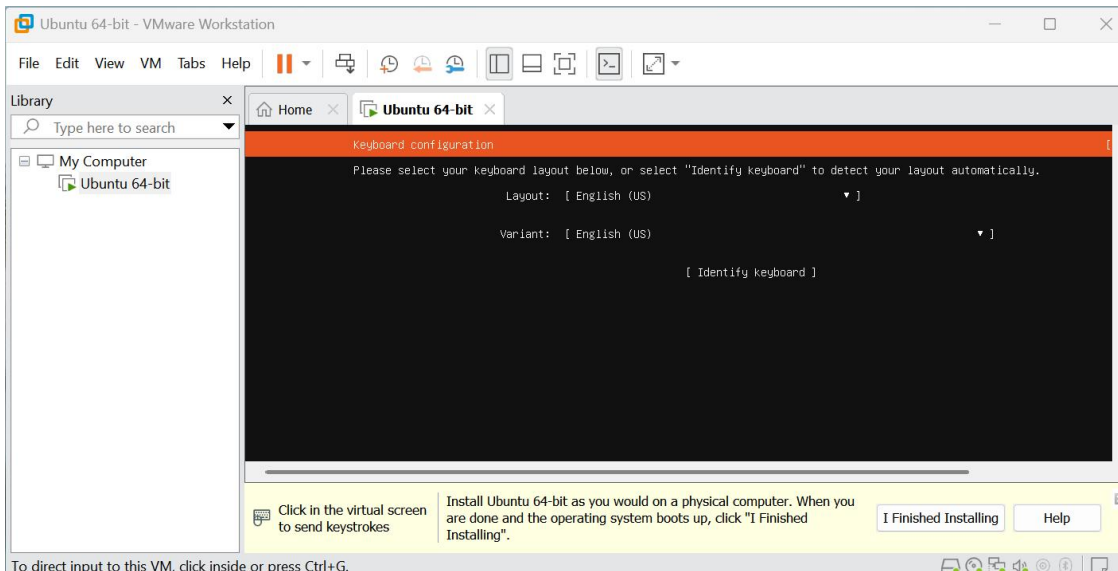


Figure 6.1: Keyboard layout and variant both set to English (US).

Task 7: Choose the Ubuntu Server Installation Type

Selected the normal Ubuntu Server installation option (not MAAS) and continued.

Task 8: Configure the Network

Step 1: Reviewed the detected network interface and confirmed it received an address through DHCP.

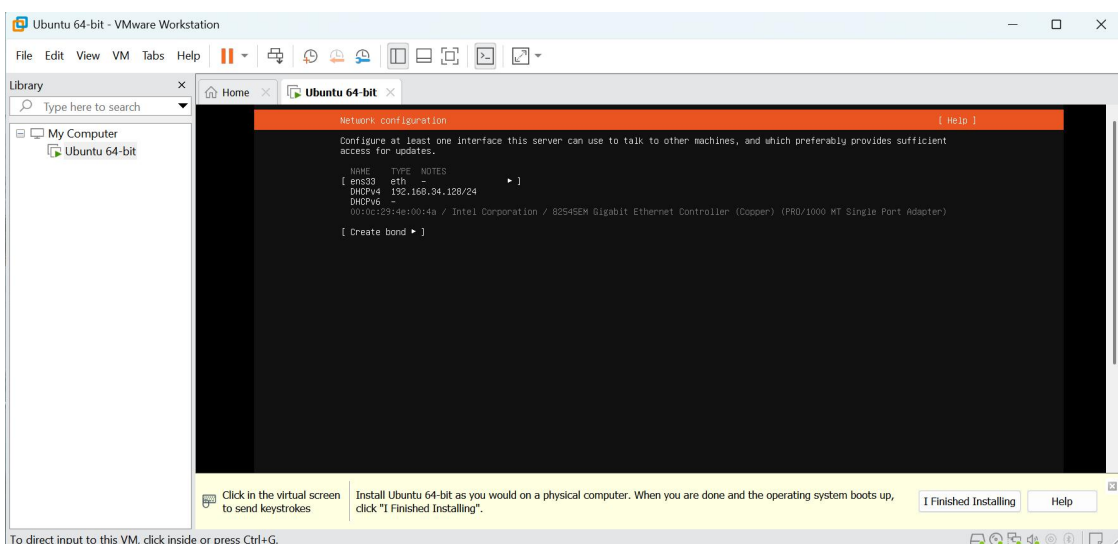


Figure 8.1: Network interface ens33 with an IPv4 address assigned by DHCP.

Step 2: Left the proxy address empty and continued.

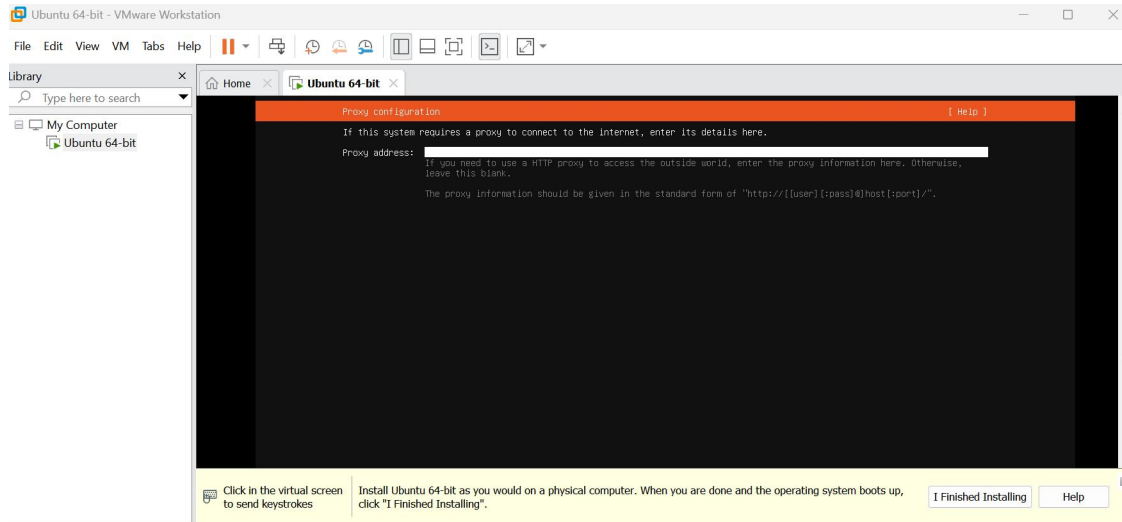


Figure 8.2: Proxy configuration screen left empty.

Step 3: Accepted the default Ubuntu archive mirror after it passed the tests.

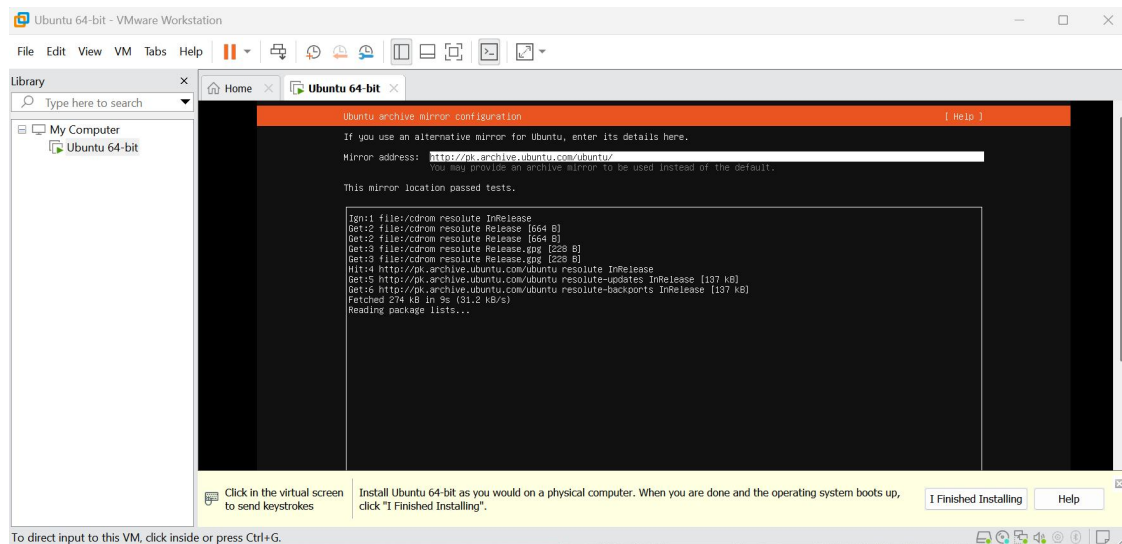


Figure 8.3: Ubuntu archive mirror configuration passed tests.

Task 9: Configure Storage

Step 1: Selected the guided storage option Use an entire disk on the VMware virtual disk.

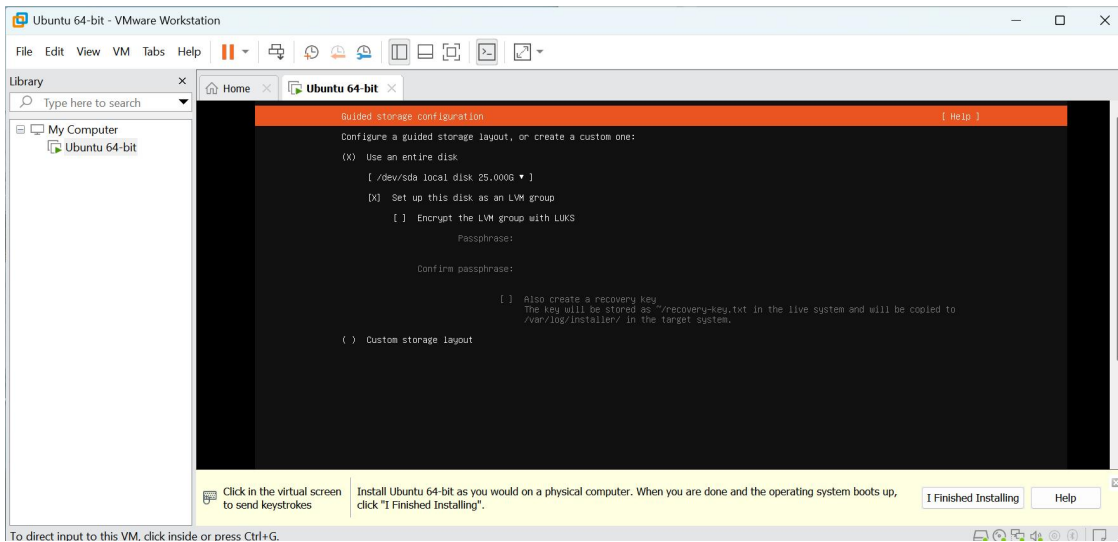


Figure 9.1: Guided storage configuration with Use an entire disk selected.

Task 10: Select the Installation Disk

Step 1: Confirmed the selected disk is the 25 GB VMware virtual disk (/dev/sda).

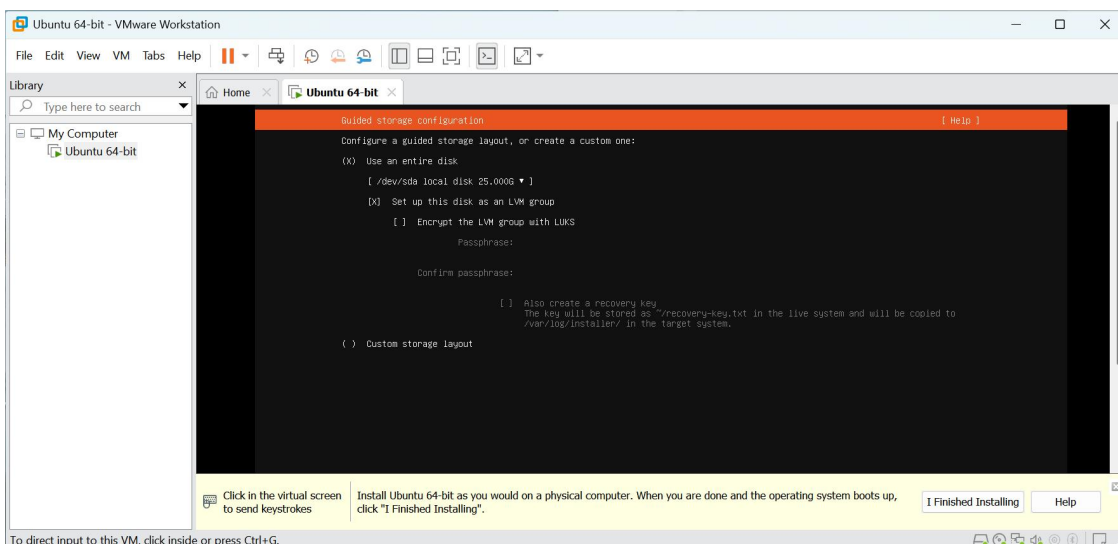


Figure 10.1: Selected disk /dev/sda local disk 25.000G.

Task 11: Review the Partitions

Step 1: Reviewed the proposed partition layout on the VMware virtual disk.

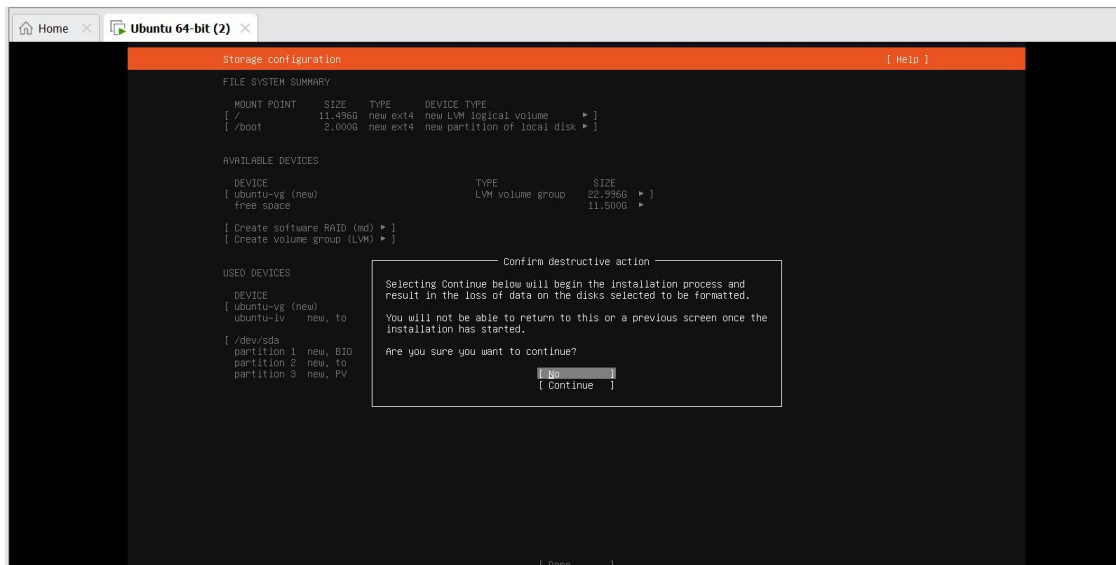


Figure 11.1: Proposed partitions and file system summary on the virtual disk.

Task 12: Confirm the Storage Changes

Step 1: Read the destructive-action warning and selected Continue after confirming the VMware virtual disk.



Figure 12.1: Confirm destructive action warning with Continue selected.

Task 13: Configure the Ubuntu Profile

Step 1: Filled in my name, the server name ubuntu, my username and a password on the profile screen.

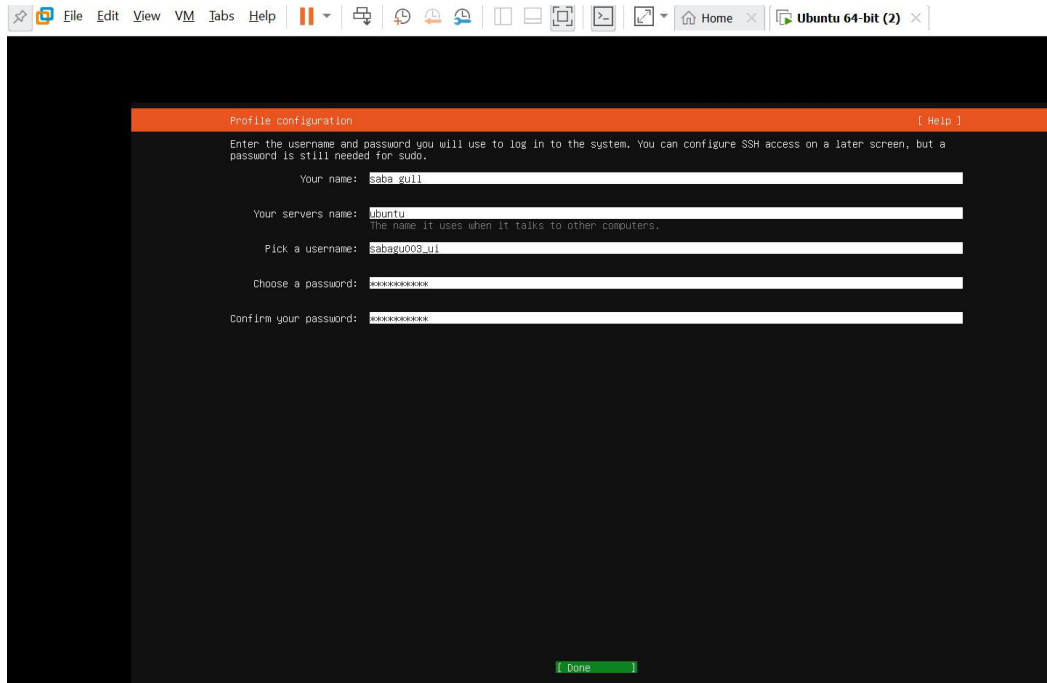


Figure 13.1: Completed Ubuntu profile configuration (password hidden).

Task 14: Complete the Software Installation

Step 1: Waited while Ubuntu Server installed the required software.

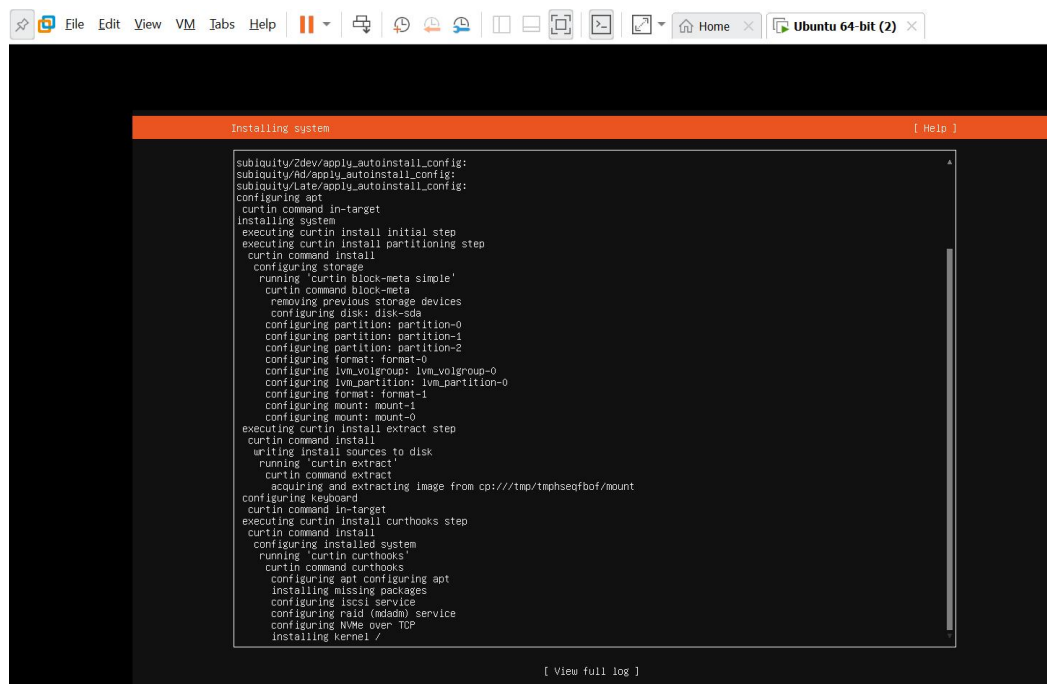


Figure 14.1: Ubuntu Server installation in progress.

Task 15: Reboot and Verify the Ubuntu Identity

Step 1: Waited for the installer to finish and selected Reboot Now.

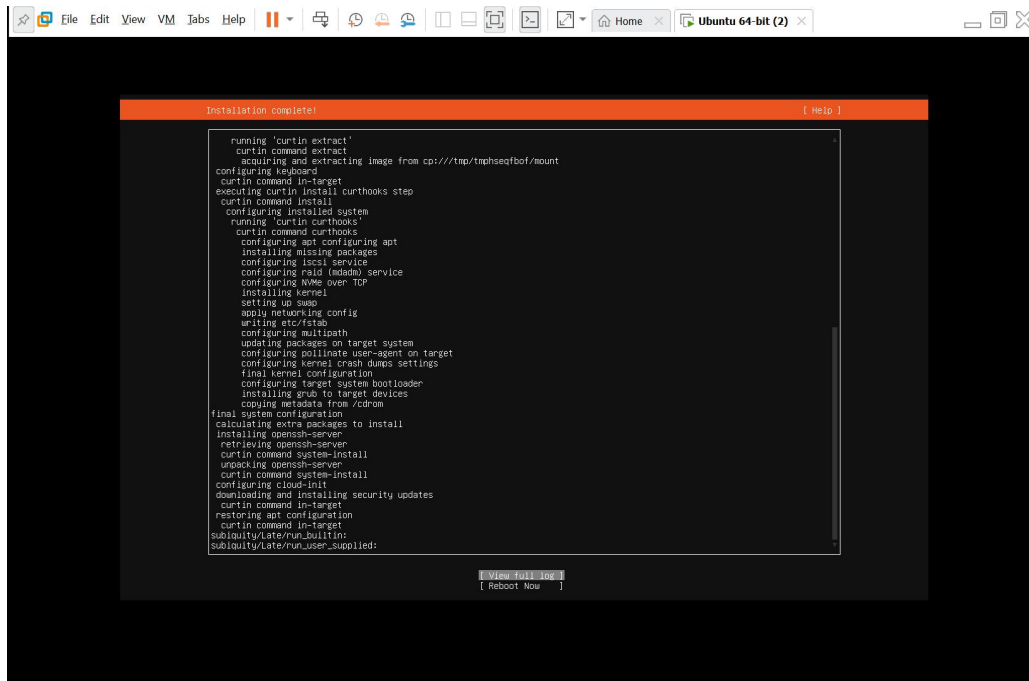


Figure 15.1: Installation complete screen with Reboot Now option.

Step 2: Rebooted, signed in with my Ubuntu username and password.

```
ubuntu login: sabagu003_ui
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Oct  4 04:17:03 PM UTC 2026

System load:  0.0          Processes:      215
Usage of /:   47.2% of 11.21GB Users logged in:    0
Memory usage: 22%         IPv4 address for ens33: 192.168.34.130
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

21 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

sabagu003_ui@ubuntu:~$
```

Figure 15.2: Successful terminal login after the reboot.

Step 3: Verified the terminal prompt follows the required format.

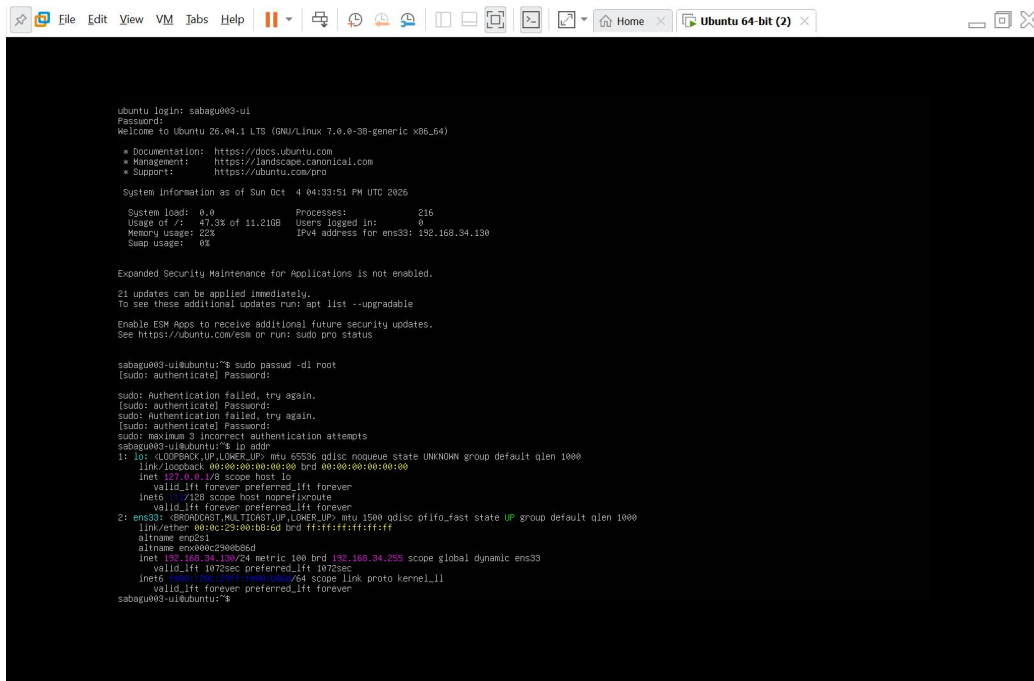
```
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status
```

```
sabagu003-ui@ubuntu:~$ |
```

Figure 15.3: Terminal prompt showing sabagu003-ui@ubuntu:~\$.

Task 16: Find the Ubuntu Server IP Address

Step 1: Ran the ip addr command and found the IPv4 address of the active interface ens33.

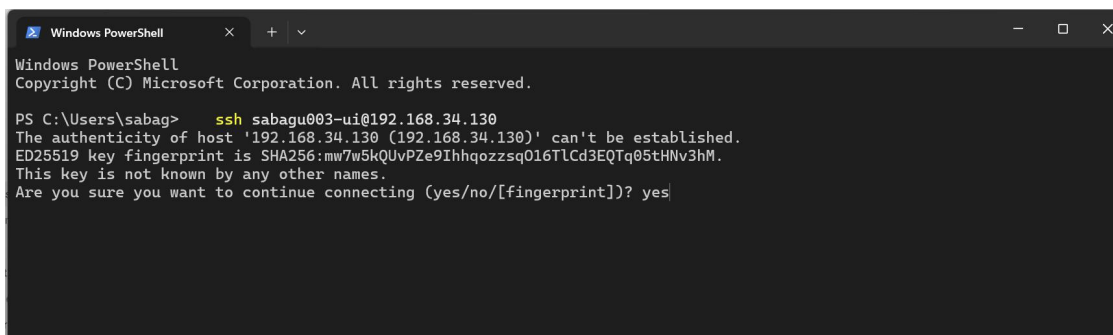


```
ubuntu login: sabagu003-ui  
Password:  
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)  
  
 * Documentation:  https://docs.ubuntu.com  
 * Management:    https://landscape.canonical.com  
 * Support:        https://ubuntu.com/pro  
  
System information as of Sun Oct 4 04:03:51 PM UTC 2026  
  
System load:  0.0          Processes:      216  
Usage of /:   47.3% of 11.21GB  Users logged in: 0  
Memory usage: 22%          IPv4 address for ens33: 192.168.34.130  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
21 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
sabagu003-ui@ubuntu:~$ sudo passwd -d root  
[sudo: authenticate] Password:  
  
sudo: authentication failed, try again.  
[sudo: authenticate] Password:  
sudo: authentication failed, try again.  
[sudo: authenticate] Password:  
sudo: maximum 3 incorrect authentication attempts  
sabagu003-ui@ubuntu:~$ ip addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
        inet 127.0.0.1/8 scope host lo  
            valid_lft forever preferred_lft forever  
        inet6 ::1/128 scope host noprefixroute  
            valid_lft forever preferred_lft forever  
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000  
    link/ether 08:00:27:00:00:00 brd ff:ff:ff:ff:ff:ff  
        altname enp0c31  
        altname ens000c30000000  
        inet 192.168.34.130/24 metric 100 brd 192.168.34.255 scope global dynamic ens33  
            valid_lft 107256s preferred_lft 107256s  
        inet6 ::7d4 scope link proto kernel_ll  
            valid_lft forever preferred_lft forever  
sabagu003-ui@ubuntu:~$
```

Figure 16.1: ip addr output showing the IPv4 address 192.168.34.130 for ens33.

Task 17: Connect to Ubuntu Server from Windows

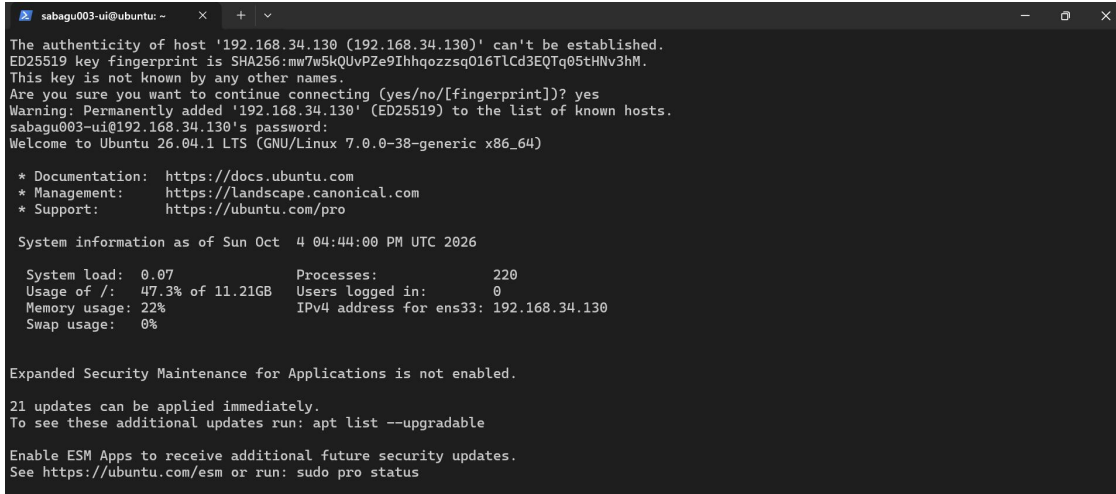
Step 1: Ran ssh sabagu003-ui@192.168.34.130 in PowerShell and typed yes to accept the fingerprint.



```
Windows PowerShell  
Copyright (C) Microsoft Corporation. All rights reserved.  
  
PS C:\Users\sabag> ssh sabagu003-ui@192.168.34.130  
The authenticity of host '192.168.34.130 (192.168.34.130)' can't be established.  
ED25519 key fingerprint is SHA256:mw7w5kQUvPZe9Ihhqozsq016TLCd3EQTq05tHNV3hM.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

Figure 17.1: SSH command and server fingerprint accepted in PowerShell.

Step 2: Entered the Ubuntu password and logged in to the server remotely.



```
sabagu003-ui@ubuntu: ~  
The authenticity of host '192.168.34.130 (192.168.34.130)' can't be established.  
ED25519 key fingerprint is SHA256:mw7w5kQUvPZe9IhhqozzsQ016TLCd3EQItQ05tHNv3hM.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.168.34.130' (ED25519) to the list of known hosts.  
sabagu003-ui@192.168.34.130's password:  
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Sun Oct  4 04:44:00 PM UTC 2026  
  
System load:  0.07          Processes:            220  
Usage of /:   47.3% of 11.21GB Users logged in:       0  
Memory usage: 22%          IPv4 address for ens33: 192.168.34.130  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
21 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status
```

Figure 17.2: Successful SSH session on the Ubuntu Server.